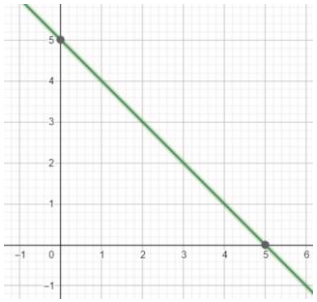
Chapter 1
Matrices and Systems of Equations

1.1 Systems of Linear Equations

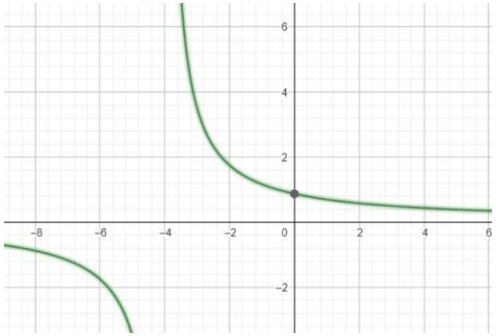
Objectives and key concepts

- Linear equations & Linear systems
  - Ax=b
  - Solution cases: consistent vs inconsistent
- Representing linear systems using matrices and vectors
  - For solutions **x** and right-hand side **b**: n-vector, n-space
  - For coefficients A: matrix
  - For linear system: augmented matrix [A|b]
- How to solve $n \times n$ linear systems
  - Elementary row operations on augmented matrices
  - Strictly triangular form: back-substitution

Linear Equations & Systems

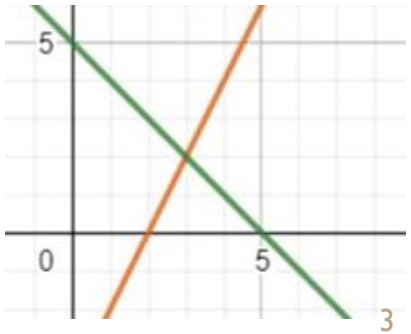


- Linear equation: $x + y = 5$
- Non-linear equation:
  - $4x_1^2 + 6x_2 = 3$
  - $2x_1x_2 + 8x_2 = 7$
  - $5x_2^{1/2} = 9$



- Linear system:

$$x + y = 5$$
$$2x - y = 4$$



Linear equations

Definition/定义

A ***linear equation（线性方程）*** in n unknowns(未知数) $x_1, x_2, \ldots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where the coefficients（系数） $a_1, a_2, \ldots, a_n$ and b are real numbers（实数）.

Linear systems/线性方程组

Definition

A ***linear system*** of m equations in n unknowns $x_1, x_2, \dots, x_n$ is a system of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \quad\quad & \quad\quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

where the coefficients a_{ij}'s and b_j's are all real numbers.

The system is an $\boldsymbol{m} \times \boldsymbol{n}$ linear system.

Goal: to solve linear systems.

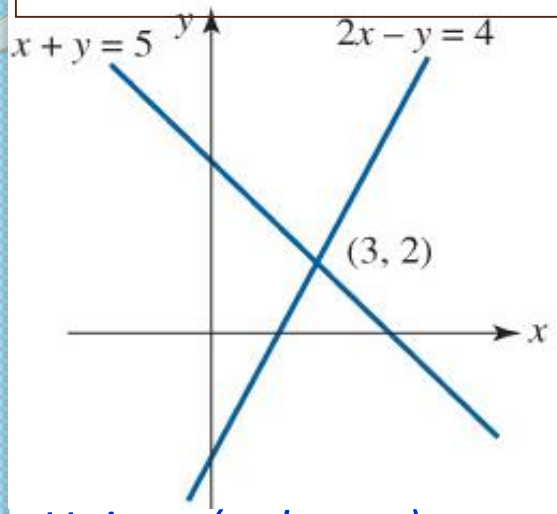
Question: What is a solution to a linear system? How many solutions are there?

Solution of linear systems

- Definition:
  A ***solution*** of an $m \times n$ system is an ordered *n-tuple of numbers*(n 元组) $x_1, x_2, \ldots, x_n$ that satisfies all the equations of the system.
  - An *inconsistent(不相容)* linear system has **no** solution;
  - A *consistent（相容）* system has **one** solution or **many** solutions.
- ***The solution set*** – the set of all solutions of a linear system.
- ***Equivalent systems*** **–** two linear systems with the same solution set.

Solution cases (2x2) 定解情况

One linear equation in 2 variables corresponds to a line in the two-dimensional (2D) plane.



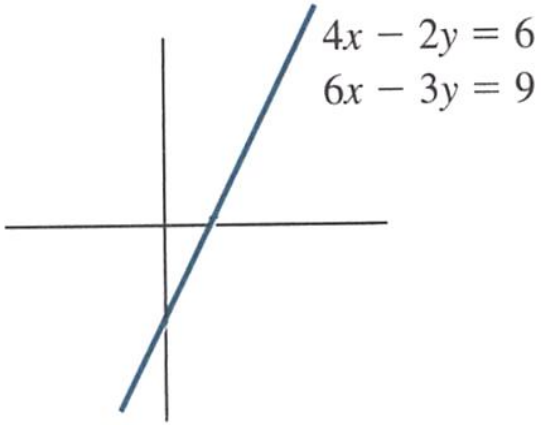
Unique (only one) solution

$$x + y = 5$$
$$2x - y = 4$$

Lines intersect at (3, 2)

Unique solution:

$$x = 3, y = 2.$$

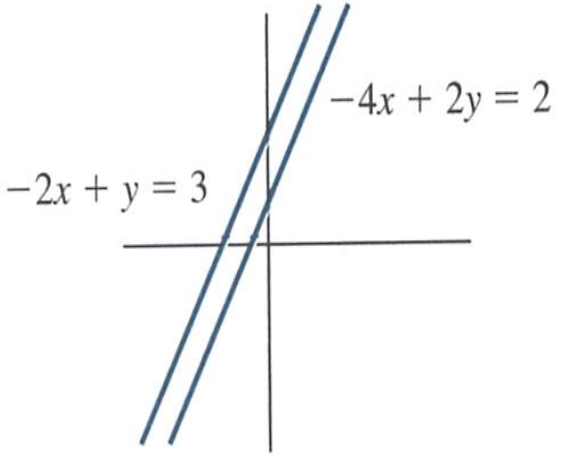


Many solutions

$$4x - 2y = 6$$
$$6x - 3y = 9$$

Both equations have the same graph. Any point on the graph is a solution.

Many solutions.



No solution

$$-2x + y = 3$$
$$-4x + 2y = 2$$

Lines are parallel.

No point of intersection

No solution.

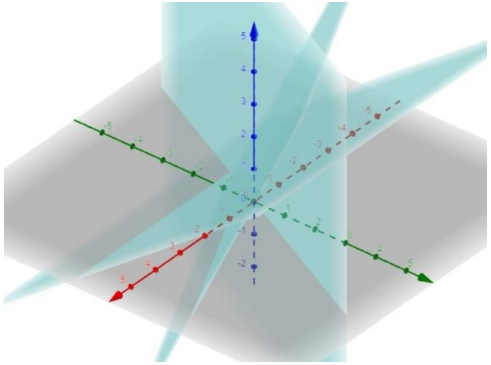
consistent (相容)

Inconsistent（不相容）

Solution cases (3x3)

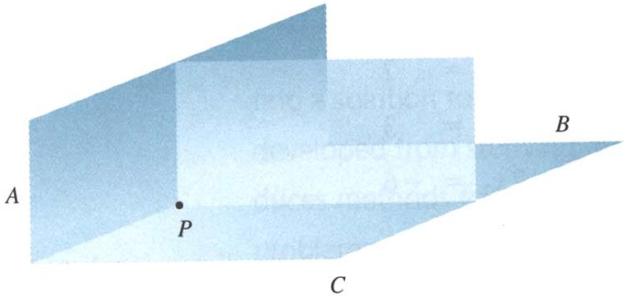
A linear equation in three variables corresponds to a plane in three-dimensional (3D) space.

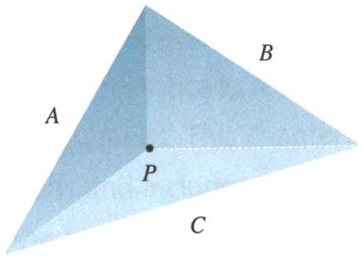
$$\begin{aligned} 2x - y &= 0 \\ -x + 2y - z &= -1 \\ -3y + 4z &= 4 \end{aligned}$$



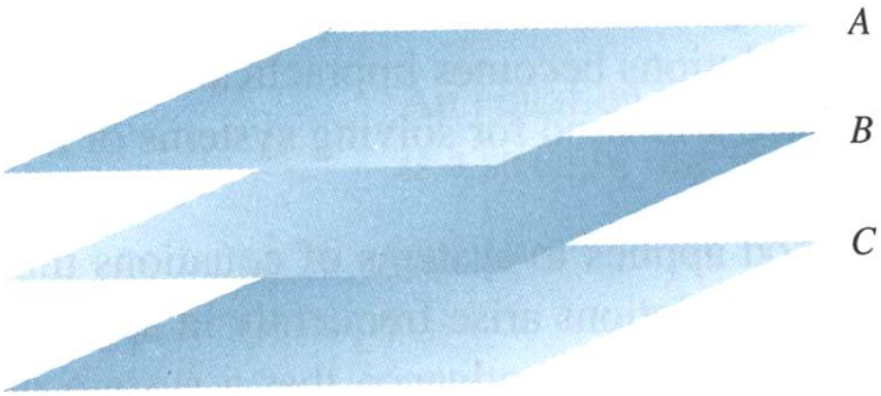
※ Systems of three linear equations in three variables:

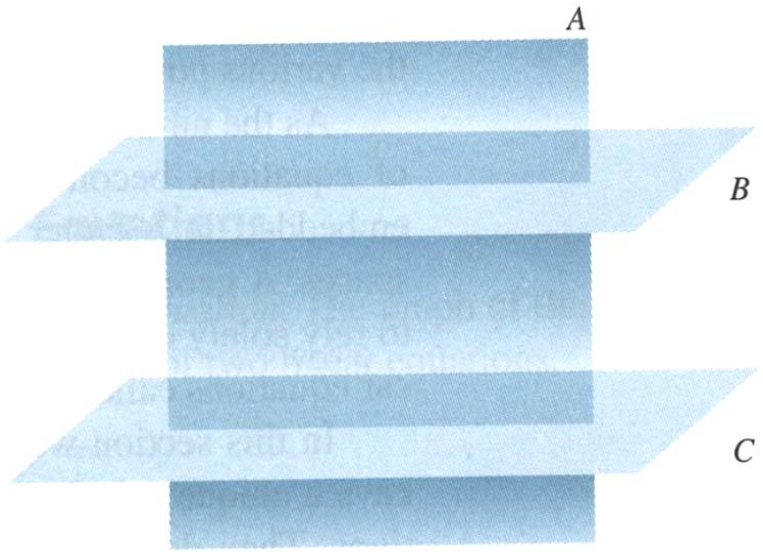
- *Unique solution*



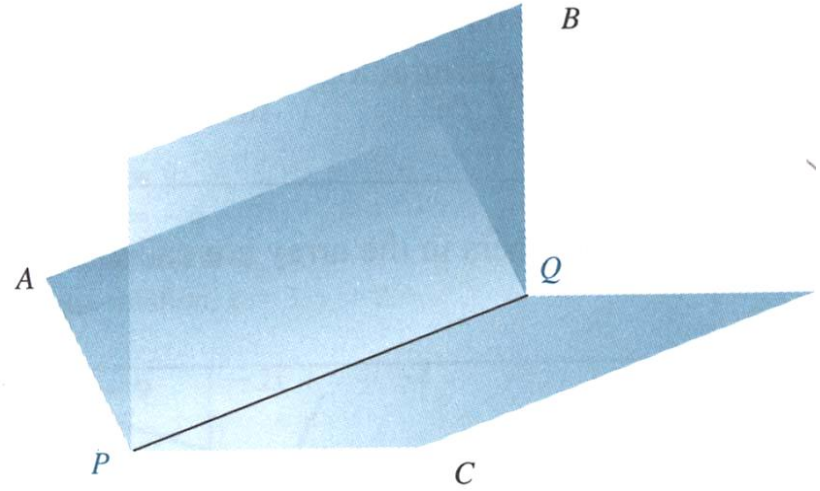


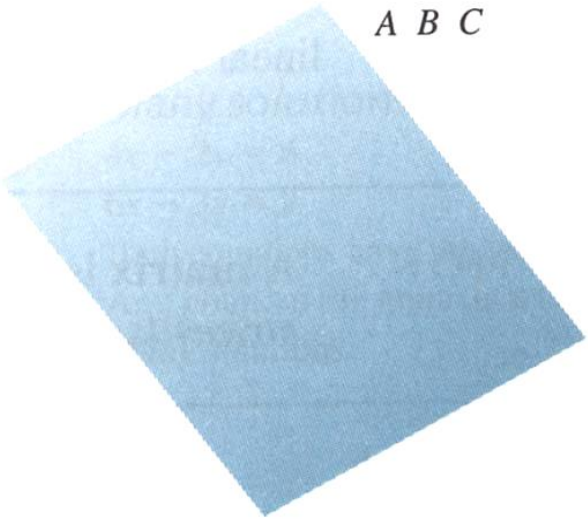
- *No solutions*





- *Many solutions*





Extend to 4D? And more?



n-dimensional real space

- Let $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$ be a tuple of n real numbers.
  - For example, geometrically(几何上讲), $\mathbb{R}^2$ is a set of all points (x_1, x_2) in the plane.
- The set of all such tuples is denoted as $\mathbb{R}^n$, called real coordinate space of dimension n.
- In linear algebra, we commonly call $\boldsymbol{x}$ in $\mathbb{R}^n$ a vector(向量), and write it vertically as a column vector:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- The right-hand side constants in a linear system also forms a vector:

$$\boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Matrix（矩阵）

Definition 矩阵　　矩形数组

- A ***matrix*** is a rectangular array of numbers.
- The numbers in the array are called the ***entries*** of the matrix.

元素

$$A=\begin{bmatrix}2 & 3 & -4\\ 7 & 5 & -1\end{bmatrix} \qquad B=\begin{bmatrix}7 & 1\\ 0 & 5\\ -8 & 3\end{bmatrix} \qquad C=\begin{bmatrix}3 & 5 & 6\\ 0 & -2 & 5\\ 8 & 9 & 12\end{bmatrix}$$

Note there are square brackets[方括号] in left and right sides of the matrix.

Matrix/矩阵, and Vector/向量

Matrices with only one row（行） or one column（列） are called n-vectors

- A $1 \times n$ matrix is a *row vector*
- An $n \times 1$ matrix is a *column vector;* $\mathbb{R}^n$

$$\begin{bmatrix} 1 & 0 & 3 \\ -2 & 4 & 5 \end{bmatrix}$$

2×3 matrix

$$\begin{bmatrix} 2 & 5 & 7 \\ -9 & 0 & 1 \\ -3 & 5 & 8 \end{bmatrix}$$

3×3 matrix

square matrix

$$\begin{bmatrix} -2 & 4 & 5 \end{bmatrix}$$

1×3 matrix

row vector

$$\begin{bmatrix} 7 \\ 1 \\ 8 \end{bmatrix}$$

3×1 matrix

column vector

Matrix, Vector

Identity Matrices （单位矩阵） 对角元素

A square matrix with diagonal value equals to 1 and other entries equal to 0, denoted as I_n,where the subscript (n) represent the size.

下标

Example

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \textit{Question: } I_4 = ?$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Row and column of a matrix

$$A = \begin{bmatrix} 2 & 3 & -4 \\ 7 & 5 & -1 \end{bmatrix}$$

$\begin{bmatrix} 2 & 3 & -4 \end{bmatrix}$ row 1

$\begin{bmatrix} 7 & 5 & -1 \end{bmatrix}$ row 2

$\begin{bmatrix} 2 \\ 7 \end{bmatrix}$ column 1

$\begin{bmatrix} 3 \\ 5 \end{bmatrix}$ column 2

$\begin{bmatrix} -4 \\ -1 \end{bmatrix}$ column 3

Matrices of a linear system (using an example)

Given the linear system

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6,$$

Organize all the coefficients in a 3x3 array of numbers:

Coefficient matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & -1 & -2 \end{bmatrix}$$

If we also write down the 3 right-hand side numbers:

Augmented matrix

增广矩阵

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{array}\right]$$

How to solve a system of linear equations?

- Strategy: replace one system with an **equivalent（等价的） system** that is easier to solve. (2x2 example)
- Consider the following two systems:

I:

$$x_1 - 2x_2 = -1$$
$$-x_1 + 3x_2 = 3$$

II:

$$x_1 - 2x_2 = -1$$
$$x_2 = 2$$

- System II. is easier to solve because it is clear from the second equation that $x_2 = 2$. Use this value in the first equation, we get $x_1 = 3$.
- System *I.* has the same solution as system *II.*, x=(3, 2).

How to solve a system of linear equations?

- Three operations that can be used on a system to obtain an equivalent system:
  - Interchange the order of two equations:

$$\begin{matrix} x+y=5 \\ 2x-y=4 \end{matrix} \sim \begin{matrix} 2x-y=4 \\ x+y=5 \end{matrix}$$

  - Multiply (divide) the same nonzero real number on both sides of equations:

$$\frac{1}{3}x+\frac{1}{3}y=\frac{2}{3} \sim x+y=2$$

  - Add to (subtract from) one equation a multiple of another equation:

$$\begin{matrix} \mathrm{x}+\mathrm{y}=5 \\ \mathrm{x}-\mathrm{y}=3 \end{matrix} \sim \begin{matrix} x+y=5 \\ 2x=8 \end{matrix}$$

- Strategy:
  - Obtain an equivalent strictly triangular form.

How to solve a system of linear equations?

Keep this example in mind:

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 4 \\ 2x_2 + 5x_3 &= 4 \\ 3x_3 &= 6 \end{aligned}$$

An $n \times n$ system is said to be in strictly triangular form if in the kth equation, the coefficients of the first $k-1$ variables are all zero, and the coefficient of x_k is nonzero.

A linear system in strictly triangular form has coefficient matrix of the following form:

$$\begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix}$$

Example

Linear System:

$$3x_1 + 2x_2 - x_3 = -2$$
$$-3x_1 - x_2 + x_3 = 5$$
$$3x_1 + 2x_2 + x_3 = 2$$

Add the first equation to the second equation:

(R2 becomes R2 + R1)

$$3x_1 + 2x_2 - x_3 = -2$$
$$x_2 = 3$$
$$3x_1 + 2x_2 + x_3 = 2$$

Subtract first equation from the third equation:

(R3 becomes R3 – R1)

$$3x_1 + 2x_2 - x_3 = -2$$
$$x_2 = 3$$
$$2x_3 = 4$$

Then, use *back substitution* to get the solution:

- From equation 3: $x_3 = 2$
- From equation 2: $x_2 = 3$
- From equation 1: $x_1 = \frac{-2-2x_2+x_3}{3} = -2$

From operations on linear systems to row operations on matrices

Linear systems correspond to augmented matrices.

For example:

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6$$

is the same as

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{array}\right]$$

Therefore, 3 operations used to obtain equivalent systems, correspond to 3 row operations on the augmented matrix.

From operations on linear systems to row operations on matrices

Elementary Row Operations: 行初等变换

- Interchange two rows.
- Multiply a row by a nonzero real number.
- Replace a row by its sum with a multiple of another row.

- **Row equivalent matrices:** Two matrices where one matrix can be transformed into the other matrix by a sequence of elementary row operations.
- **Fact about Row Equivalence:** If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

Notation

$\overset{\sim}{R1+(-3)R2}$

Add (-3)× R2 to R1, hence R1 is changed but R2 keep intact.

For example:

(1) $\left[\begin{array}{ccc|c} 1 & -2 & 4 & 12 \\ 2 & -1 & 5 & 18 \\ -1 & 3 & -3 & -8 \end{array}\right] \overset{\sim}{R1+2R2} \left[\begin{array}{ccc|c} 5 & -4 & 14 & 48 \\ 2 & -1 & 5 & 18 \\ -1 & 3 & -3 & -8 \end{array}\right]$

(2) $\left[\begin{array}{ccc|c} 1 & -2 & 4 & 12 \\ 2 & -1 & 5 & 18 \\ -1 & 3 & -3 & -8 \end{array}\right] \overset{\sim}{R2+2R1} \left[\begin{array}{ccc|c} 1 & -2 & 4 & 12 \\ 4 & -5 & 13 & 42 \\ -1 & 3 & -3 & -8 \end{array}\right]$

Example

Solving the following system of linear equation

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6$$

Solution

Equation Method

Initial system:

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6$$

$$x_1 + x_2 + x_3 = 2$$

Eq2+(–2)Eq1 $\quad x_2 - x_3 = -1$

Eq3+(–1)Eq1 $\quad -2x_2 - 3x_3 = -8$

Matrix Method

Augmented matrix:

Pivotal row

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{array}\right]$$

$$\sim \begin{array}{l} \\ \text{R2+(–2)R1} \\ \text{R3+(–1)R1} \end{array} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & -3 & -8 \end{array}\right]$$

$$x_1 + x_2 + x_3 = 2$$
$$x_2 - x_3 = -1$$
$$-2x_2 - 3x_3 = -8$$

Pivot row

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & -3 & -8 \end{bmatrix}$$

Eq3+(2)Eq2

$$x_1 + x_2 + x_3 = 2$$
$$x_2 - x_3 = -1$$
$$-5x_3 = -10$$

$\sim$ R3+(2)R2

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -5 & -10 \end{bmatrix}$$

The solution is

$x_1 = -1, x_2 = 1, x_3 = 2$

The solution is

$x_1 = -1, x_2 = 1, x_3 = 2$

Check! Is this a solution of the original system?

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